MGI Raw Data Interpretation & Demultiplexing

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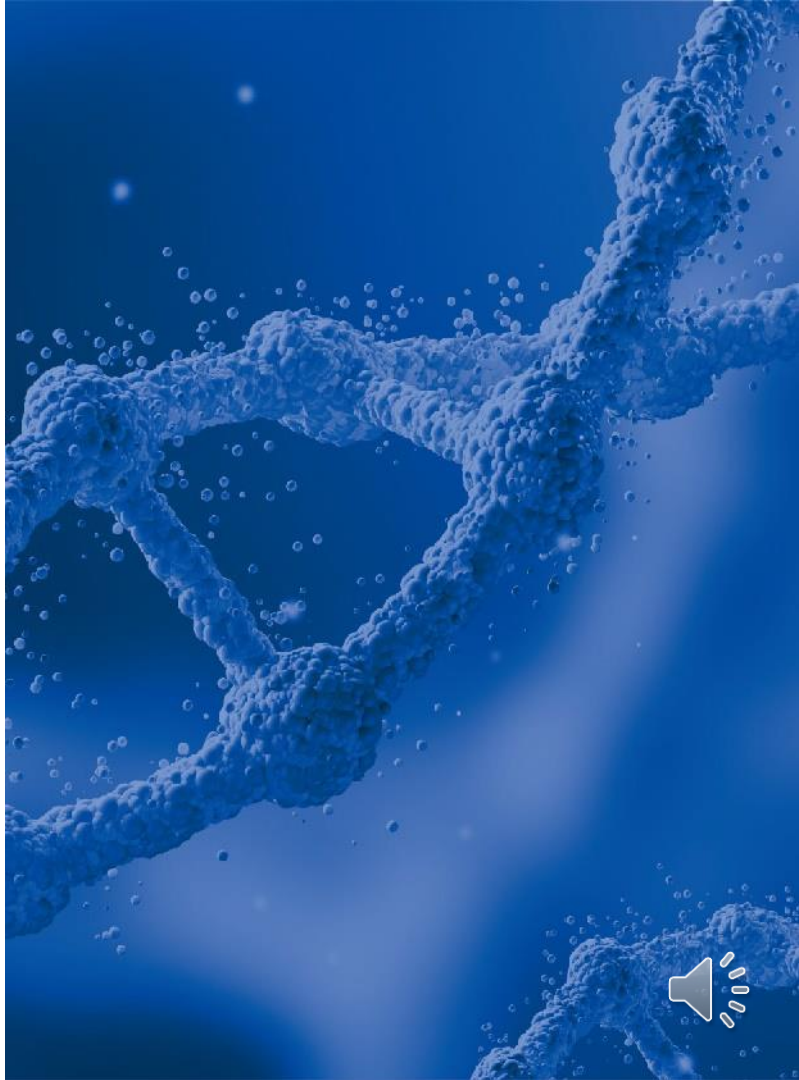

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MPS Raw Data Interpretation

Raw Data Transmission

MGISEQ sequencer

DNB → FastQ, CAL

Bioinformatics analysis server

- Raw FastQ → Clean FastQ + report
- CAL → FastQ

Raw Data

```
@S100002265L1C001R003378873/1
ACCTTTGTGAATTCAAGAGTCAGGATATGGAGC
+
AHICCCIDFAADDHA@IAI@F@IF5CACIIA=I
```

Check report、FastQ

Raw Data Interpretation (MGISEQ-200)

```
S300000686/L01
├── BarcodeStat.txt *
├── S300000686_L01_10_1.fq.fqStat.txt
├── S300000686_L01_10_1.fq.gz
├── S300000686_L01_10_2.fq.fqStat.txt
├── S300000686_L01_10_2.fq.gz
├── S300000686_L01_discard_1.fq.fqStat.txt
├── S300000686_L01_discard_1.fq.gz
├── S300000686_L01_discard_2.fq.fqStat.txt
├── S300000686_L01_discard_2.fq.gz
├── S300000686_L01_ambiguous_1.fq.gz
├── S300000686_L01_ambiguous_2.fq.fqStat.txt
├── S300000686_L01_ambiguous_2.fq.gz
├── S300000686_L01_undecoded_1.fq.fqStat.txt
├── S300000686_L01_undecoded_1.fq.gz
├── S300000686_L01_undecoded_2.fq.fqStat.txt
├── S300000686_L01_undecoded_2.fq.gz
└── SequenceStat.txt *
```

Windows

V300029737_L02_1_1.fq	2019/9/15 5:46	好压 GZ 压缩文件	3,669,936...
V300029737_L02_1_2.fq	2019/9/15 5:46	好压 GZ 压缩文件	3,643,189...
V300029737_L02_2_1.fq	2019/9/15 5:46	好压 GZ 压缩文件	3,776,331...
V300029737_L02_2_2.fq	2019/9/15 5:46	好压 GZ 压缩文件	3,737,343...
V300029737_L02_3_1.fq	2019/9/15 5:46	好压 GZ 压缩文件	3,386,012...
V300029737_L02_3_2.fq	2019/9/15 5:46	好压 GZ 压缩文件	3,374,994...
V300029737_L02_4_1.fq	2019/9/15 5:46	好压 GZ 压缩文件	2,931,656...
V300029737_L02_4_2.fq	2019/9/15 5:46	好压 GZ 压缩文件	2,920,292...
V300029737_L02_97_1.fq	2019/9/15 5:46	好压 GZ 压缩文件	5,512,143...
V300029737_L02_97_2.fq	2019/9/15 5:46	好压 GZ 压缩文件	5,471,759...
V300029737_L02_98_1.fq	2019/9/15 5:46	好压 GZ 压缩文件	6,266,180...
V300029737_L02_98_2.fq	2019/9/15 5:46	好压 GZ 压缩文件	6,229,568...
V300029737_L02_99_1.fq	2019/9/15 5:46	好压 GZ 压缩文件	6,426,088...
V300029737_L02_99_2.fq	2019/9/15 5:46	好压 GZ 压缩文件	6,429,663...
V300029737_L02_100_1.fq	2019/9/15 5:46	好压 GZ 压缩文件	5,719,174...
V300029737_L02_100_2.fq	2019/9/15 5:46	好压 GZ 压缩文件	5,691,499...
V300029737_L02_101_1.fq	2019/9/15 5:46	好压 GZ 压缩文件	6,107,550...
V300029737_L02_101_2.fq	2019/9/15 5:46	好压 GZ 压缩文件	6,072,579...
V300029737_L02_102_1.fq	2019/9/15 5:46	好压 GZ 压缩文件	7,733,724...
V300029737_L02_102_2.fq	2019/9/15 5:46	好压 GZ 压缩文件	7,391,595...
V300029737_L02_103_1.fq	2019/9/15 5:47	好压 GZ 压缩文件	7,224,475...
V300029737_L02_103_2.fq	2019/9/15 5:47	好压 GZ 压缩文件	7,167,548...
V300029737_L02_104_1.fq	2019/9/15 5:47	好压 GZ 压缩文件	7,110,895...
V300029737_L02_104_2.fq	2019/9/15 5:47	好压 GZ 压缩文件	7,051,079...

File Naming (MGISEQ-200)

- S300000686_L01_**10**_1.fq.gz

"**FC**"_"**Lane**"_"**Barcode**"_"**Read1/2**".fq.gz
FastQ file for subsequent Bioinformatics analysis
If sequenced Pair-end (PE), Read2 FastQ file would also be generated.

- S300000686_L01_**undecoded**_1.fq.gz

"**FC**"_"**Lane**"_"**unmatch Barcode**"_"**Read1/2**".fq.gz
FastQ file for reads whose barcode does not match the barcode index used in sequencing during demultiplexing process.

- S300000686_L01_**ambiguous**_1.fq.gz

"**FC**"_"**Lane**"_"**Same Barcode**"_"**Read1/2**".fq.gz
FastQ file for reads that has more than 2 mismatches in barcode index during demultiplexing process.

- S300000686_L01_**discard**_1.fq.gz

"**FC**"_"**Lane**"_"**filter by basecall**"_"**Read1/2**".fq.gz
FastQ file for reads that are filtered out during base-calling process.

Document interpretation（MGISEQ-200）

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'BarcodeStat.txt' Document Interpretation

#Barcode	Correct	Corrected	Total	Percentage(%)
barcode1	10, 009, 720	7, 436, 788	17, 446, 508	5.405185
barcode2	16, 032, 135	1, 625, 814	17, 657, 949	5.470692
barcode4	27, 112, 689	8, 866, 046	35, 978, 735	11.146741
barcode5	26, 980, 226	1, 235, 013	28, 215, 239	8.741496
barcode6	30, 062, 251	1, 269, 224	31, 331, 475	9.706952
barcode7	16, 066, 233	8, 646, 233	24, 712, 466	7.656285
barcode8	36, 399, 032	1, 121, 131	37, 520, 163	11.624298
barcode9	31, 441, 899	1, 068, 190	32, 510, 089	10.072104
barcode10	30, 508, 029	1, 267, 831	31, 775, 860	9.844628
barcode11	18, 579, 900	8, 854, 725	27, 434, 625	8.499650
barcode12	25, 244, 371	715, 615	25, 959, 986	8.042786
Barcode No.	Barcode without mismatch reads number (SE)	Barcode with mismatch reads number (SE)	Barcode with mismatch + without mismatch reads number (SE)	Barcode total reads demultiplexed/ Raw data total reads
Total	268, 691, 349	42, 179, 291	310, 870, 640	96.312294

Document interpretation(MGISEQ-200)

'SequenceStat.txt' Document Interpretation

#Sequence	Barcode	Count	Percentage(%)
TTCCTCCT	barcode8	36, 399, 032	11.276955
GCTGTAAG	barcode9	31, 441, 899	9.741163
GACATCTC	barcode10	30, 508, 029	9.451836
ACCTCTTC	barcode6	30, 062, 251	9.313727
TCCTGGTA	barcode4	27, 112, 689	8.399910
CATTGACG	barcode5	26, 980, 226	8.358871
TGCGATAG	barcode12	25, 244, 371	7.821077
CAACCGTA	barcode11	18, 579, 900	5.756326
CATTCGTC	barcode7	16, 066, 233	4.977555
TAGGAGCT	barcode2	16, 032, 135	4.966991
CCGATGTA	barcode1	10, 009, 720	3.101158
CAACCGGA	barcode11	7, 844, 890	2.430462
TCCTGGGA	barcode4	7, 741, 528	2.398439
CATTCGGC	barcode7	7, 355, 278	2.278773
CCGATGGA	barcode1	6, 875, 337	2.130081
TAGGAGGT	barcode2	1, 068, 780	0.331124
Barcode index	Barcode NO.	Reads per Barcode	Reads per Barcode / Barcode total Reads

FastQ file format

MGI FastQ format

```
@S200013802L1C001R00300000091/1
TTATCAGGCCTGGGCAGTGTTAGAACCGTAGGTCGGATAAGGCGTTTACGCCGCATCCGACACGCATTGCACGATGCCGCAAAGGCATAAAAAGTCGA
+
@DEDF?FFF?@FFFFEFDF@@?FE?FFFD?FF@FFF?@@?FFFF@@D?FFFFFFE@FFF?F?FFFEDAFFEFFE@FFFFFE@EFFFE@EE??EF@FFEDFFF
```

Illumina FastQ format

Illumina generates raw data files are in binary base call (BCL) format, **bcl2fastq** is included, standalone conversion software that demultiplexes data and converts BCL files to standard FASTQ file formats for downstream analysis.

```
@A00251:152:HYJNCDSXX:1:1101:1199:1000 1:N:0:GTAGTGCA+TGCACTAC
ATTTCGAATGATAAAGTGGTAATACAGGCTATGGGTAAAATTTGATAGGCATCTGAGGAGGGAGACATTATTTCAGGCTGGAAAGAGAATTGTCTTTAGAACTGCCCATT
+
FFFFFFFFFFFFFFFF,FFFFFFFFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFFFFFF
```

A

F

Line 1: read id, begins with @
Line 2: sequences
Line 3: +
Line 4: base quality (Phred 33)

'F' ASCII value is 70, therefore the quality of base 'A' equals to 70-33=37

Illumina vs MGI Read ID Format

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Illumina

@ A00251 : 152 : HYJNCDSXX : 1 : 1101 : 1199: 1000 1 : N : 0 : GTAGTGCA+TGCACTAC

- Read ID Identifier
- Machine serial number
- Run ID
- Flowcell ID
- Lane number
- Tile number (same as FOV)
- Tile X axis and Y axis location
- Read1 (1) or Read2 (2)
- Unfiltered set Y, and others set N
- When none of the control bits are on
- Barcode sequence

MGI

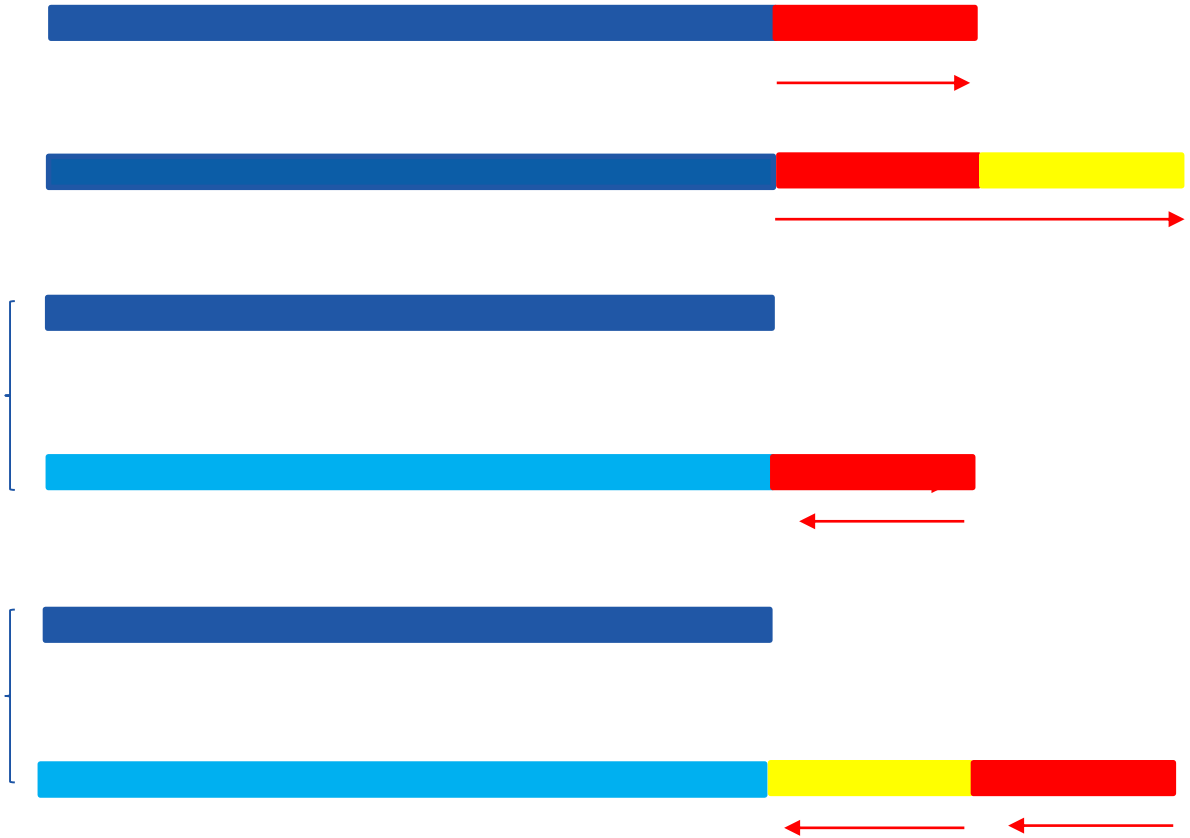
@ S200013802 L1 C001R0030 0000091 /1

- Read ID Identifier
- Flowcell ID
- Lane number
- FOV (same as tile number)
- Read number
- Read1 (1) or Read2 (2)

2

MGI Demultiplexing

MGI FastQ file sequence structure



Read1

Read2

Barcode1

Barcode2

MGI Demultiplexing– Sequencer demultiplexing

Demultiplexing: (PE150 sequencing, single barcode, barcode length 10bp, two correction bases)

C: \ZebracallV2\1.0.8.208>client.exe D: \Fakepath\V30073166\L01 312 6 72 –P 151 –N V30073166 –U 1 –F –E 3 –B D: \Fakepath\Barcode.csv --barcode_start 301 --barcode_len 10 --barcode_mismatch 2

- CAL Path : D: \Fakepath\V30073166\L01
- Barcode path: D: \Fakepath\Barcode.csv
- Cycle number : 312
- Start col : 1
- Start row : 1
- End col : 6
- End row : 72

Barcode length / bp	Barcode mismatch / bp (MGI recommendation)
6	0
8	1
10	2

Basecall lite download link: https: //pan.genomics.cn/ucdisk/s/JjAJza
Zebracall download link: https: //pan.genomics.cn/ucdisk/s/jieljq

MGI Demultiplexing — SplitBarcode

① PE150 sequencing, single barcode, length 8bp, two correction bases, CAL file as input

```
[shengqin@cngb-login-0-21 bin]$ ./splitBarcode -F V300070185/L01/calFile -C 310 --Col 6 --Row 72 -P 151 -N V300070185 -U 1 -E 3 -o R100400180009 -B barcode.list -i 301 8 1 -r -t 30 -m 100
```

② PE150 sequencing, single barcode, length 8bp, FastQ file as input

```
[shengqin@cngb-login-0-21 bin]$ ./splitBarcode -B barcode.list -1 V300063151/L01/V300063151_L01_read_1.fq.gz -2 V300063151/L01/V300063151_L01_read_2.fq.gz -b 300 8 1 -r -o result -t 30 -m 100
```

Basecall lite download link: https: //pan.genomics.cn/ucdisk/s/JjAJza
Zebracall download link: https: //pan.genomics.cn/ucdisk/s/jieljq

Thanks for your time!

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